



FACULTY OF AGRICULTURE, ENGINEERING & NATURAL SCIENCES

SCHOOL OF ENGINEERING AND THE BUILT ENVIRONMENT

I3582IM

ENGINEERING MATHEMATICS II

Study Guide
2024

Dr L. Matsebula

Copyright ©¹ University of Namibia,
Namibia.

Printed and published by the University of Namibia

¹©All rights reserved. Apart from any fair dealing for the purpose of research, criticism, or review as permitted under the Copyright Act 98 of 1978, no part of this material may be reproduced, stored in a retrieval system, transmitted or used in any form, or be published, redistributed, or screened by any means electronic, photocopying, recording, or otherwise without the prior written permission of the University of Namibia

Administrative details

Welcome

Welcome to I3582IM – ENGINEERING MATHEMATICS II. In this module, you will develop competence in various *mathematical techniques*, and learn how to apply these techniques to the various fields of engineering.

Contact information

The following are the contact details for the lecturers for this module:

Name:	Dr L. Matsebula	(Course coordinator)
Office:	CV125	
Tel no:	065 232 4247	
Email:	lmatsebula@unam.na	

Name:	Mr S. Amunime	(Tutor)
Office:	CV115	
Tel no:	065 232 4237	
Email:	samunime@unam.na	

Complaint procedure

If you have any concerns regarding this module or the teaching of this module, please direct your concerns to the Course Coordinator: Dr L. Matsebula (lmatsebula@unam.na)

If you are still dissatisfied, please direct your concerns to the Head of the Department: Dr Ester Angula (eangula@unam.na)

It is very important that you follow this order when directing your concerns. Suppose you direct your concerns outside of this order (or to any other person), in that case, your email will be redirected to follow this hierarchy (which will only delay your concerns being addressed).

The subject line of your written (via email) concern should be: Complaint: I3582IM. Your email should include full detail regarding your concerns, or it will be sent back to you to request clear and precise information.

When contacting a lecturer or administrator by email please include the following:

- A polite greeting
- The course code (I3582IM)
- Your query
- Your name and student number at the end of the email
- Your title – i.e. whether you are Ms, Miss, Mrs, Mr, Dr etc.

Consultation

The following times are the consultation times for the lecturer. Consultation outside of these times can be arranged via email or in person.

Lecturer	Day	Time
Dr L. Matsebula	Tuesday	10H00-12H00

Textbooks

The recommended textbooks for I3582IM are

- Ross S.L. (1991). Introduction to Ordinary Differential Equations, 4th Edition, John Wiley and Sons.
- Kreyszig, E. (2020). Advanced Engineering Mathematics. 10 Edition, John Wiley and Sons.
- Bird, J. (2017). Engineering Mathematics. 8th Edition, Routledge.
- Stroud, K.A. and Booth, D.J.(2020). Engineering Mathematics, 8th Edition, Red Globe Press.
- Stewart, J., Clegg, D.K., and Watson, S, (2020). Calculus: Early Transcendentals. 9th Edition, Cengage Learning.

Lectures and Tutorials

Lectures

The lectures and tutorials are conducted in person. Attend all your lectures and tutorials for the module. There are two lectures for the module each week. See details below.

- Session 1

Day: Monday

Time: 10H30-12H30

Venue: Auditorium

- Session 2

Day: Tuesday

Time: 08H00-10H00

Venue: Auditorium

Tutorials

The saying “mathematics is not a spectator sport” should always be remembered. You will not learn if you only attend lectures. We learn mathematics by solving problems.

Each week a *list of problems* will be posted on Moodle. Your work (in addition to attending lectures) is to do as many of these problems as possible. We learn very little by watching someone else solve mathematics problems. We learn only when we struggle through solving problems by ourselves.

If you need feedback on attempted problems, you can contact a tutor or a lecturer (via email). We will attempt to get an answer back to you within *72 hours* (assuming we are not flooded with queries).

Every second week you will also have a *Tutorial test* and or a Moodle quiz to complete. There will be *no sick tests* for *missed* tutorial tests. The *marks* for these assessments will contribute towards your *continuous assessment mark*, which in turn contributes to your *final mark*.

Note that tutorial attendance is compulsory, and a register will be taken.

- Session 1

Day: Thursday

Time: 08H00-10H00

Venue: Auditorium

Module information

Moodle

Moodle will be used for:

- Communicating administrative information on the course.
- Important announcements.
- Distributing lecture notes.
- Collaborate sessions.

You should log on to Moodle *daily*.

Module details

Module name	I3582IM
Module NQF level	5
NQF Credits	12
Duration of module	1 Semester (12 weeks)

Table 1: Course details

Purpose of the module

The purpose of this module is to introduce students to the fundamentals of engineering mathematics and its applications to science and engineering. The primary focus of this module is the study of *Matrix algebra*, *Integration techniques and their applications*, *Differential equations*, *Functions*, *Integral Transforms*, *Fourier Series and Transforms*, and *Polar coordinates and graphs*.

Upon completion of this module, students should be able to:

1. Calculate eigenvalues and eigenvectors and relate them to Engineering solutions,
2. Solve calculus problems using various integration techniques,

3. Apply calculus to trigonometric functions to solve mathematical and engineering problems,
4. Solve engineering problems using 1^{st} order and 2^{nd} order differential equations,
5. Define and analyze Fourier series of real-valued functions.
6. Use Laplace and Fourier transforms in solving differential equations.

Description of the module

The content of this module is taken from the recommended list of textbooks.

- Chapter 1: Further integration
 - Further integration techniques
 - 1.1 Integration by u -substitution
 - 1.2 Integration by parts
 - 1.3 Reduction formula
 - 1.4 Trigonometric substitution
 - Applications of the definite integral:
 - 1.5 Areas under the curves
 - 1.6 Volumes of Revolution
- Chapter 2: Differential Equations
 - 1^{st} ORDER ODEs
 - 2.1 Meaning and solutions
 - 2.2 Linear Ordinary Differential Equations
 - 2.3 Separable Ordinary Differential Equations
 - 2.4 Homogeneous Ordinary Differential Equations
 - 2.5 Exact Ordinary Differential Equations
 - Solution to 2^{nd} Order ODEs
 - 2.6 Method of undetermined coefficients
 - 2.7 Method of variation of parameters
- Chapter 3: Integral Transforms
 - Integral Transforms
 - 3.1 Laplace Transforms
 - 3.2 Laplace Transforms of derivatives and integrals
 - 3.3 Inverse Transforms of derivatives and integrals
 - 3.4 Unit step functions
 - 3.5 Application to solve 1^{st} and 2^{nd} ordinary differential equations
- Chapter 4: Fourier Series and Transforms
 - Fourier Series and Transforms
 - 4.1 Fourier series
 - 4.2 Fourier sine and cosine series
 - 4.3 Applications in boundary value problems
- Chapter 5: Further Matrix Algebra
 - Further Matrix Algebra
 - 5.1 Eigenvalues and eigenvectors
 - 5.2 Hermitian and unitary matrices
 - 5.3 Quadratic forms and change of axes
- Chapter 6: Functions
 - Limits and continuity of functions
 - 6.1 Limit at a point
 - 6.2 Improper limit and continuity
 - 6.3 Exponential functions
 - 6.4 Logarithmic functions
 - 6.5 Hyperbolic functions
- Chapter 7: Polar coordinates/Graphs
 - Polar coordinates/Graphs
 - 7.1 Definition of polar coordinates
 - 7.2 Relationship between Cartesian and polar coordinates

Semester programme

We will follow the following programme during the semester. Minor changes may be made to it, thus, make sure that you always have the latest copy of this document (see Table 2). Should any changes be made to it, communication will be made to that effect.

Table 2: Course Outline

Week	Day	Date	Lecture Content	TUT Test Scope
1	Monday	29 July	1.1	
	Tuesday	30 July	1.2	
2	Monday	05 August	1.3	
	Tuesday	06 August	1.4	
3	Monday	12 August	1.5	1.1—1.3
	Tuesday	13 August	1.6	
4	Monday	19 August	2.1	
	Tuesday	20 August	2.2	
Thursday 22 AUGUST SEMESTER TEST 1				
MID SEMESTER BREAK 26 AUGUST-31 AUGUST				
5	Monday	02 September	2.3	
	Tuesday	03 September	2.4	
6	Monday	09 September	2.5	
	Tuesday	10 September	2.6	
7	Monday	16 September	2.7	2.1—2.7
	Tuesday	17 September	3.1	
8	Monday	23 September	3.2	
	Tuesday	24 September	3.3	
Thursday 26 SEPTEMBER SEMESTER TEST 2				
9	Monday	30 September	3.4	
	Tuesday	01 October	3.5	
10	Monday	07 October	4.1	3.1—3.5(Online)
	Tuesday	08 October	4.2	
11	Monday	14 October	4.3	
	Tuesday	15 October	5.1	
12	Monday	21 October	5.2	
	Tuesday	22 October	5.3	
13	Monday	28 October	Revision	4.1—4.3, 5.1—5.3 (Online)
	Tuesday	29 October	Revision	

Gantt Chart

The second semester timelines are outlined in a Gantt chart in Figure 1.

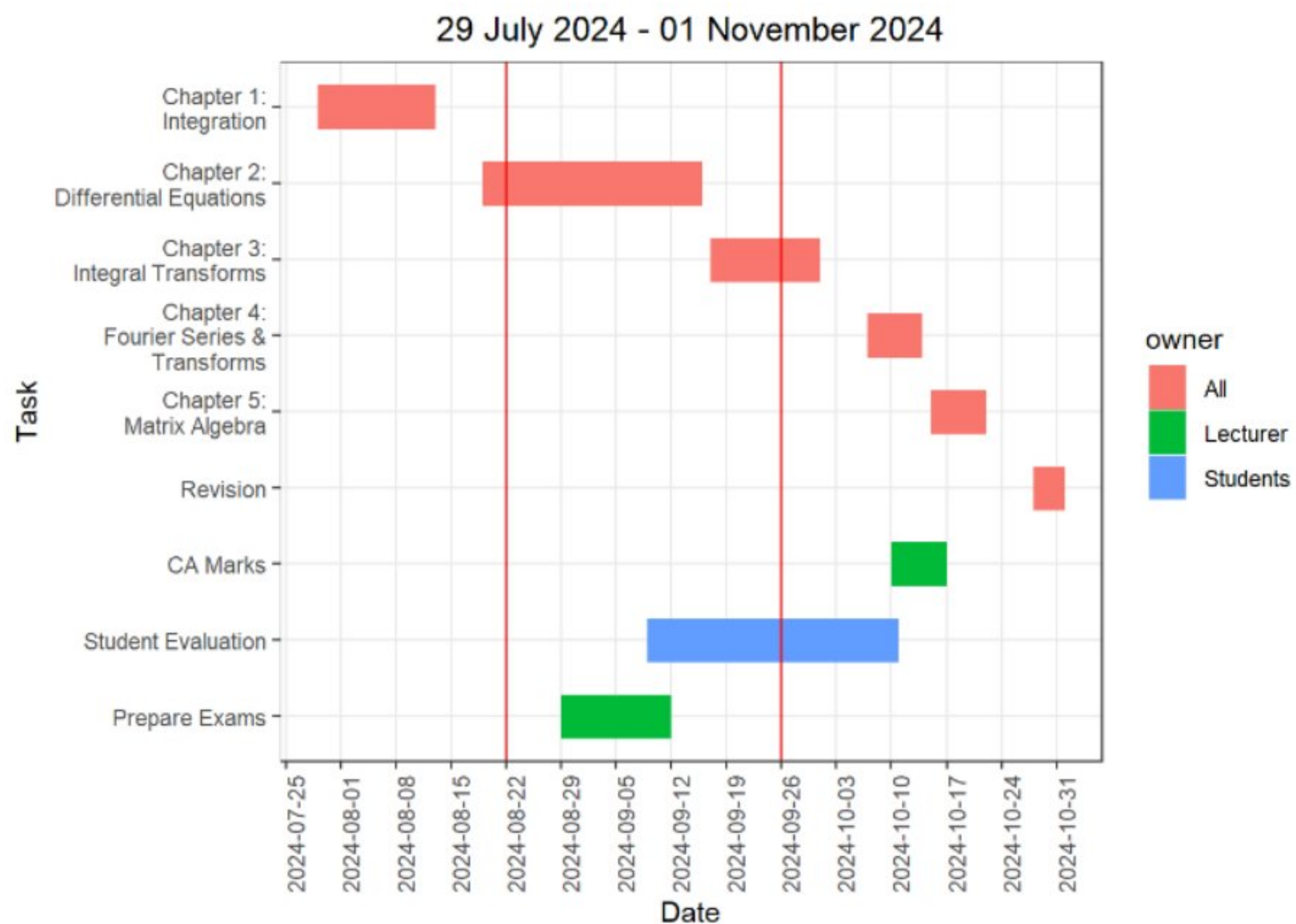


Figure 1: Engineering Mathematics II Gantt chart. The solid red lines denote the semester test dates. Both students and the lecturer are responsible for completing all tasks highlighted in red. The tasks marked in blue are only for students, whilst those in green are only for the lecturer.

Assessments

Note that all assessments are to be held on campus, and you must make sure that you avail yourself on the dates listed in the schedule and assessment table (See Table 3).

The dates for the semester tests are highlighted in the semester programme. We will summarise the contents of each semester test here.

Table 3: Assessment details

Test	Scope	Date	Time	Venues
Semester Test 1	Chapter 1	22/08/2024	08.00—10.00	Auditorium
Semester Test 2	Chapter 2	26/09/2024	08.00—10.00	Auditorium
Sick Test	See Test 1 & 2	TBA	TBA	TBA

The scope of the examination is the entire course content.

Procedure for application for sick test

An *application request* for a special assessment opportunity (sick test) must be submitted to the lecturer via *Moodle within 7 calendar days* after the assessment has been written. The *outcome* of your application will be communicated to you via *Moodle*. Valid reasons for applying to write a special assessment include:

- *Being sick* to the extent that you are not able to write (the original medical certificate must be included as proof); or
- *Prior application* (with confirmation from the sports bureau or cultural office) for participation in provincial, national or international sports or cultural events; or
- The *death* of a close family member or friend (a copy of the death certificate or funeral letter must be included as proof); or
- Other excuses deemed acceptable by the lecturer.

Students who have already *started* with an assessment opportunity or *signed the attendance register* may *not* apply for a special assessment opportunity. No special assessment opportunity will be granted for missed special assessment opportunities. This applies to semester tests only.

Passing the module

To pass the module, you must, first of all, qualify to write the examination. This is achieved by getting a minimum of 40% for your Continuous Assessment (CA). The CA mark (SM/FPM), will be made up of the following:

$$\text{CA} = 40\% \text{ Tutorial Tests average} + 30\% \text{ Semester Test 1} + 30\% \text{ Semester Test 2.}$$

If your CA mark is less than 40%, you have automatically failed the module.
The Final Mark (FM) will be calculated as follows:

$$\text{Final Mark} = 50\% \text{ CA Mark (SM)} + 50\% \text{ Examination Mark (EM)}.$$

You are deemed to have passed if the following marks have been obtained:

- last summative assessment mark (Examination Mark) of at least 40 %; and
- a final mark of at least 50 %. (This means that if you obtain a final mark of 50 % but have not met the 40 % requirement for the examination mark, you will fail the module. You will qualify for a supplementary examination in such a case).

Supplementary Examination

The Assessment Committee or a senior administrative officer of a faculty/college (in this case Faculty of Engineering) in which the module resides may grant a student a supplementary last summative assessment (Supplementary Examination) opportunity if:

- (a) the student failed a module but obtained a final mark of at least 40 %; or
- (b) the student failed a module but obtained a semester/year mark of at least 60%.

The final mark of the module is capped at 50%. Note that the continuous assessment mark is still used for the final mark calculation. Observe also that the department plays no role in the granting of supplementary examinations.

Special last summative assessment/examination

Special summative assessment opportunities are considered by the faculty/college in which the programme/qualification resides, for students who, in the event of illness, for compassionate reasons, on religious grounds or for similar legitimate reasons, were prevented from attending a summative assessment opportunity. Students may be granted a special summative assessment opportunity if they apply for it within seven calendar days after the original date of the relevant summative assessment opportunity. The Executive Dean or his/her delegated authority, in consultation with the relevant HOD, considers all applications and decides whether or not to grant the special summative assessment opportunity. The Faculty/College Board determines the procedure for and manner of such application in accordance with University procedure. Note that the application is to be done at the Faculty of Engineering. Again the department plays no role here.

The Assessment Committee or the Executive Dean or his/her delegated authority of a faculty/college in which the qualification resides may grant a student a supplementary last summative assessment opportunity, if the student requires not more than the equivalent of two semester modules or one year module for the completion of the relevant qualification provided that the student:

- (a) was registered for the relevant module in the current academic year; and
- (b) was admitted to, and participated in the last assessment opportunity in the relevant module; and

- (c) has complied with all the experiential or practical requirements prescribed for the qualification (where applicable), excluding work-integrated modules. The Executive Dean or his/her delegated authority of the faculty/college in which the qualification resides may, in exceptional circumstances and in consultation with the Executive Dean of the faculty/college, in which the particular modules reside, waive one or more of the conditions specified in (c).

Academic Integrity

Any form of academic dishonesty is taken extremely seriously by the University. The following is an extract from the Student Ethics and Judicial Services summary:

Students should take special notice of the importance with which the University regards academic integrity. A student who is found guilty of a transgression relating to academic dishonesty (for example possessing notes during a test or examination or plagiarising an assignment) may face a penalty that ranges from cancellation of the module in which the transgression was committed to permanent expulsion from the University.

Plagiarism

Students must make themselves aware of the definition of plagiarism. The full University plagiarism policy can be found in the full set of regulations above.